Handheld-gripper data on other robot hands: where we stand, and what our lab should try first

Lab meeting, October 2026

Haokai Ding

Handheld-gripper data is scalable, but the embodiment gap blocks direct reuse.

UMI-style interfaces separate the teacher from the robot and replay demos directly[2]; FastUMI scales this[1]. Transfer fails when the executing end-effector differs from the handheld gripper[2][5].

FastUMI standardizes the interface and replaces GoPro VIO with a RealSense T265.

ISO-standard flange plates and adjustable camera mounts keep viewpoints consistent[1]; plug-in fingertips for five common grippers cover >90% of Open X-Embodiment grippers[1][3]; T265 removes complex calibration and coordinate transforms[1].

FastUMI's 10k+ trajectories show depth and data scale improve transfer-relevant policies.

>10,000 real-world demos span 22 tasks, 19 object categories, 12 skills, ~6–12 s[1]; depth raises Diffusion Policy success 53.33→80.00% Pick Lid and 20.00→93.33% Open Ricecooker[1]; Pick Cup success 20.00/26.67/53.33% with 200/400/800 demos[1].

A different end-effector breaks geometry, actions, tracking, and mechanics.

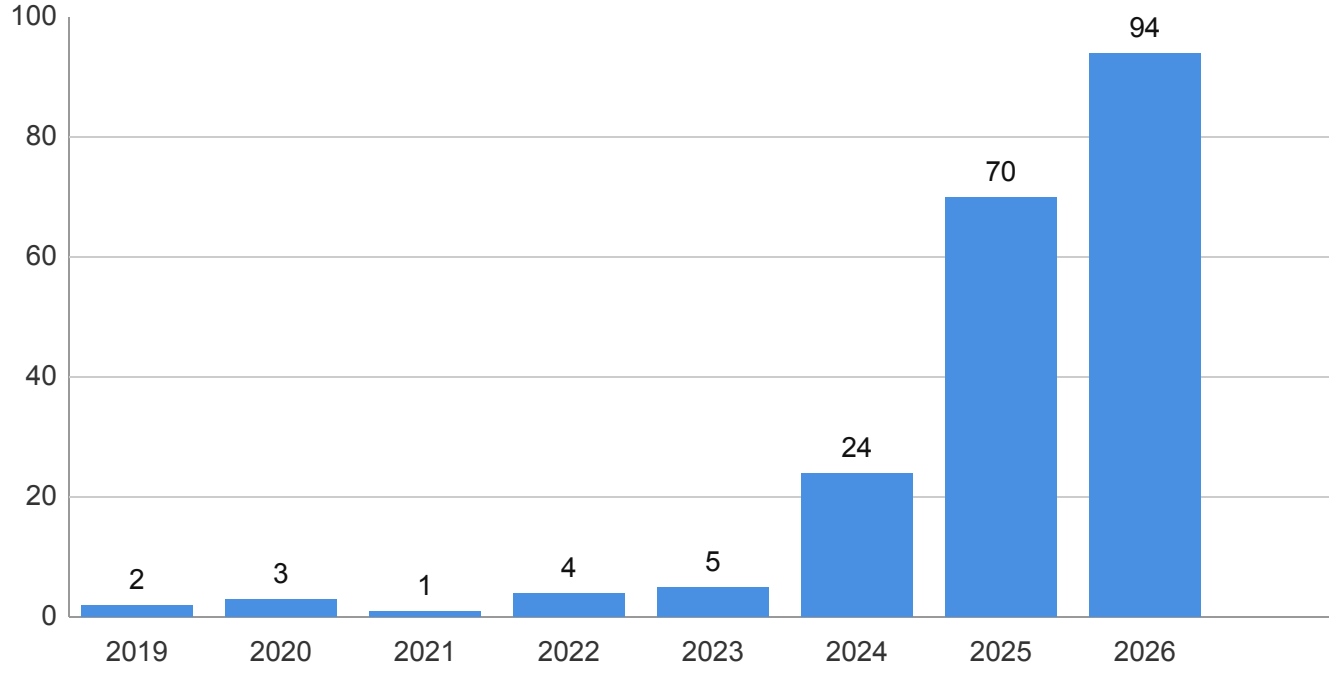
Grasp geometry and contact points shift with shape and kinematics[5]; Diffusion Policy relative TCP poses do not by themselves handle embodiment[11]; T265 degrades under close-range occlusion and low light and may need reinitialization[1]; non-parallel jaws displace TCP, corrected by local-Z compensation[1].

The literature offers pooling, retargeting, policy/data fixes, and useful contrasts.

- Pool across embodiments: Open X-Embodiment[3]; RT-1[8].
- Retarget mismatched hands: Doshi[5]; Sanches[6]; Dexcap[7].
- Fix policy/data: Diffusion Policy[11]; Data Quality[12].
- Contrasts: UMI[2]; human video[4]; VR teleop[9]; Grasping in the Wild[10].

The field is very recent: 80.8% of 203 papers appeared in 2025 or later.

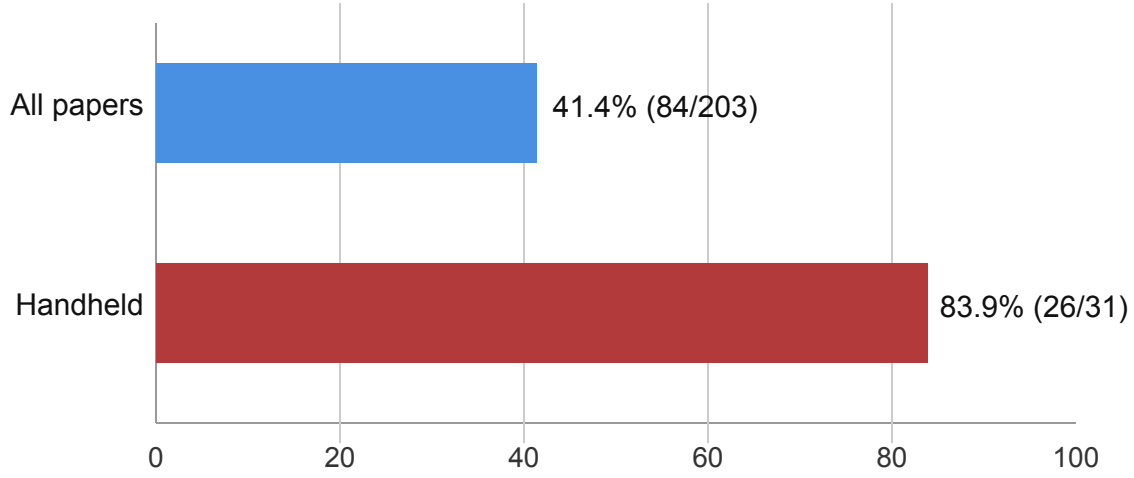
Counts rose from 24 papers in 2024 to 70 in 2025 and 94 in 2026 through October[14].



2026 is partial through October; labels were produced by a language model from abstracts, with a relevance spot check of 19/20[14].

Handheld grippers are robot-free far more often than the field, but still a minority.

Handheld: 31 papers, 26 robot-free (83.9%); all: 84/203 robot-free (41.4%). Only 16.7% release datasets; 33.0% address cross-embodiment[14].



Only 27 papers state dataset size (median 2,100 demos, range 3–1,200,000); 108 abstracts do not specify an end-effector; some dexterous-hand devices are labelled other_teleop, so 31 is a lower bound[14].

Open problems: data quality, unsupported grippers, mismatched evaluation, sensing, and policy-first fixes.

- Re-annotate occluded/close-range recordings[1][12].
- Retarget one unsupported gripper and measure the drop[1][5][6][7].
- Evaluate mismatched end-effectors with per-task success[3][8][14].
- Test tactile/force or wireless, and try depth/data first[1][11].

Approve a 4-week pilot: clean data, build one retargeting baseline, evaluate mismatched transfer.

Week 1: clean/re-annotate occluded handheld recordings[1][12]. Weeks 2–3: retarget one gripper outside FastUMI's five attachments[1][5][6][7]. Week 4: Diffusion Policy on mismatched gripper; report per-task success drop[3][11][14]. Need: one unsupported gripper, compute, existing data.

References

1. Zhaxizhuoma et al., FastUMI, 2025, arXiv:2409.19499.
2. Cheng Chi et al., Universal Manipulation Interface, 2024, arXiv:2402.10329.
3. Open X-Embodiment Collaboration, Open X-Embodiment, 2024, arXiv:2310.08864.
4. Bahl, Gupta & Pathak, Human-to-robot imitation in the wild, 2022, arXiv:2207.09450.
5. Doshi, Huang & Coros, On hand-held grippers and the morphological gap, 2023, arXiv:2311.01832.
6. Sanches et al., Scalable intuitive human-to-robot skill transfer with wearable HMIs, 2023, DOI:10.1109/iros55552.2023.10341661.
7. Wang et al., Dexcap, 2024, arXiv:2403.07788.
8. Brohan et al., RT-1, 2022, arXiv:2212.06817.
9. Zhang et al., Deep imitation learning from VR teleoperation, 2018, arXiv:1710.04615.
10. Song et al., Grasping in the Wild, 2020, DOI:10.1109/lra.2020.3004787.
11. Cheng Chi et al., Diffusion Policy, 2023, arXiv:2303.04137.
12. Belkhale, Cui & Sadigh, Data Quality in Imitation Learning, 2023, arXiv:2306.02437.
13. Survey dataset of 203 arXiv papers on demonstration-collection systems, 2019–Oct 2026, compiled for this review.